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CSE - Shift - I (2nd Year - 2023-24)

MAIT ROHINI

Progg. In Java (2023-24) (4th Sem)
(Complete Java File)

Programming In Java LAB

(CIC-258)

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Semester : 4th Semester



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MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

Computer Science & Engineering Department

VISION

"To be centre of excellence in education, research and technology transfer in the field of computer engineering and promote entrepreneurship and ethical values."

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To foster an open, multidisciplinary and highly collaborative research environment for producing world-class engineers capable of providing innovative solutions to real-life problems and fulfil societal needs.

LAB Assessment Sheet

S.No .	Experiment	M A R K S					Total	Date of	Date of	Signature
	Name	R1	R2	R3	R4	R5	Marks	Perf.	Check.	
1.	Welcome_Message							01-02-24	08-02-24	
2.	ASCII Code of character							01-02-24	08-02-24	
3.	Sum of 2 Integers							01-02-24	08-02-24	
4.	Swap (using Bitwise)							01-02-24	08-02-24	
5.	Character_Order							08-02-24	15-02-24	
6.	Colour_Code							08-02-24	15-02-24	
7.	Even_Numbers							08-02-24	15-02-24	
8.	Floyds_Format							08-02-24	15-02-24	
9.	Palindrome_Check							08-02-24	15-02-24	
10.	ASCII to Char							15-02-24	22-02-24	
11.	Reverse 2D Array							15-02-24	22-02-24	
12.	Stack using LL							15-02-24	22-02-24	
13.	Queue using LL							15-02-24	22-02-24	
14.	Tokenizer							15-02-24	22-02-24	
15.	Area_Calculator							15-02-24	22-02-24	
16.	ATM Machine Code							15-02-24	22-02-24	
17.	Box_Volume							22-02-24	29-02-24	
18.	'This' use							22-02-24	29-02-24	
19.	Instance_Counter							22-02-24	29-02-24	
20.	Cube_Calculator							22-02-24	29-02-24	
21.	Method_Overriding							22-02-24	29-02-24	
22.	Simple_Inheritance							22-02-24	29-02-24	
23.	MultiLevel_Inheritance							22-02-24	29-02-24	
24.	'super' Keyword							22-02-24	29-02-24	
25.	Dynamic Polymorphism & Interface Overriding							22-02-24	29-02-24	
26.	Abstract Class In Java							29-02-24	07-03-24	
27.	Packages In Java							29-02-24	07-03-24	
28.	Application Of Interface							29-02-24	07-03-24	
29.	Problem On Packages							29-02-24	07-03-24	

Program – 01.

Q :- WAP To Input A String And Print The Input String?

```
import java.util.*;

public class Prog01{

    public static void main(String args[]){

        Scanner sc=new Scanner(System.in);

        System.out.print("ENTER YOUR NAME ::: ");

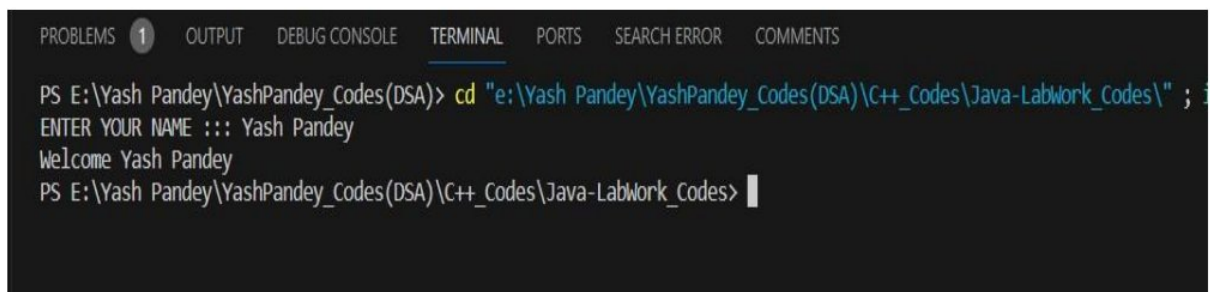
        String Name=sc.nextLine();

        System.out.println("Welcome "+Name);

    }

}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ; java Prog01
ENTER YOUR NAME ::: Yash Pandey
Welcome Yash Pandey
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes> |
```

Program – 02.

Q :- WAP To Input A Char And Print It's ASCII Value?

```
import java.util.*;

public class Prog02 {

    public static void main(String args[]){

        Scanner sc=new Scanner(System.in);

        System.out.print("ENTER YOUR CHARACTER ::: ");

        char Char=sc.nextLine().charAt(0);

        System.out.println("THE ASCII VALUE OF CHAR

            "+Char+" IS ::: "+(int)Char);

    }

}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
ENTER YOUR CHARACTER ::: a
THE ASCII VALUE OF CHAR a IS ::: 97
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes> |
```

Program – 03.

Q :- WAP To Input 2 Int Value And Display Their Sum ?

```
import java.util.*;

public class Prog03 {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("ENTER THE FIRST NUMBER ::: ");

        int a = sc.nextInt();

        System.out.print("ENTER THE Second NUMBER ::: ");

        int b = sc.nextInt();

        System.out.println("THE SUM OF " + a + " AND " +

            b + " IS ::: " + (a + b));

    }

}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
ENTER THE FIRST NUMBER ::: 3
ENTER THE SECOND NUMBER ::: 4
THE SUM OF 3 AND 4 IS ::: 7
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes> |
```


Program – 04.

Q :- WAP To Swap 2 Number Using Bitwise Operater?

```
import java.util.*;

public class Prog04 {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("ENTER THE FIRST NUMBER ::: ");

        int a = sc.nextInt();

        System.out.print("ENTER THE SECOND NUMBER ::: ");

        int b = sc.nextInt();

        System.out.println("Before Swap ::: THE Value Of a="
                               + a + " And b=" + b);

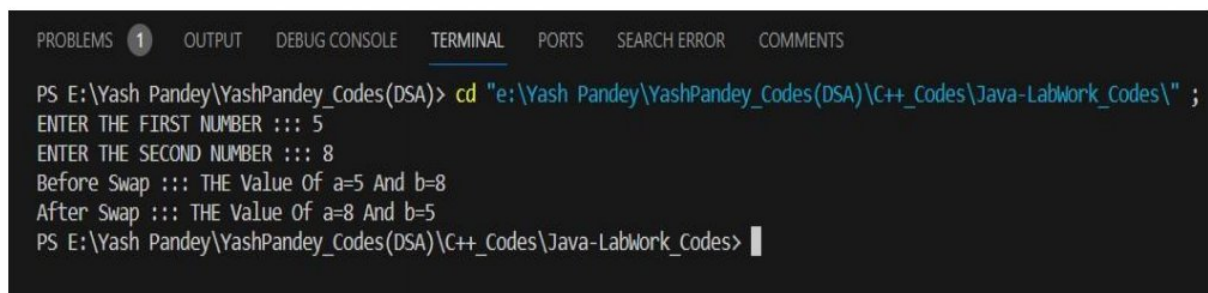
        a = a ^ b;          b = a ^ b;          a = a ^ b;

        System.out.println("After Swap ::: THE Value Of a=" +
                               a + " And b=" + b);

    }

}
```

} OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\" ;
ENTER THE FIRST NUMBER ::: 5
ENTER THE SECOND NUMBER ::: 8
Before Swap ::: THE Value Of a=5 And b=8
After Swap ::: THE Value Of a=8 And b=5
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes> |
```

Program – 05.

Q :- WAP To Display 2 Char In Alphabetical Order?

```
import java.util.*;

public class Prog05 {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("ENTER YOUR FIRST CHARACTER ::: ");

        char Char1 = sc.nextLine().charAt(0);

        System.out.print("ENTER YOUR SECOND CHARACTER ::: ");

        char Char2 = sc.nextLine().charAt(0);

        if (Char1 > Char2) {

            System.out.println(""" + Char1 + "" > "" + Char2 + "");

        } else

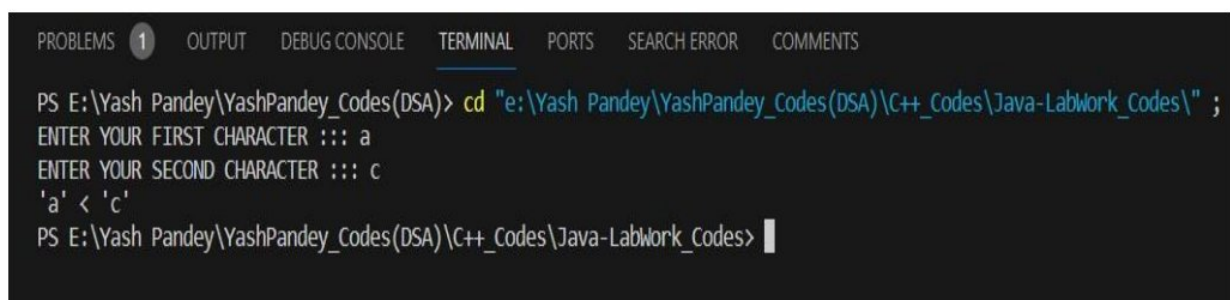
            System.out.println(""" + Char1 + "" < "" + Char2 + "");

        }

    }

}
```

OUTPUT : -

A screenshot of a code editor window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', 'PORTS', 'SEARCH ERROR', and 'COMMENTS'. The 'TERMINAL' tab is active. The terminal shows the following text: 'PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;', 'ENTER YOUR FIRST CHARACTER ::: a', 'ENTER YOUR SECOND CHARACTER ::: c', and ''a' < 'c''. The prompt 'PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>' is shown at the bottom with a cursor.

Program – 06.

Q :- WAP To Input Colour code As char from the User?

```
import java.util.*;

public class Prog06 {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("ENTER YOUR COLOUR CODE ::: ");

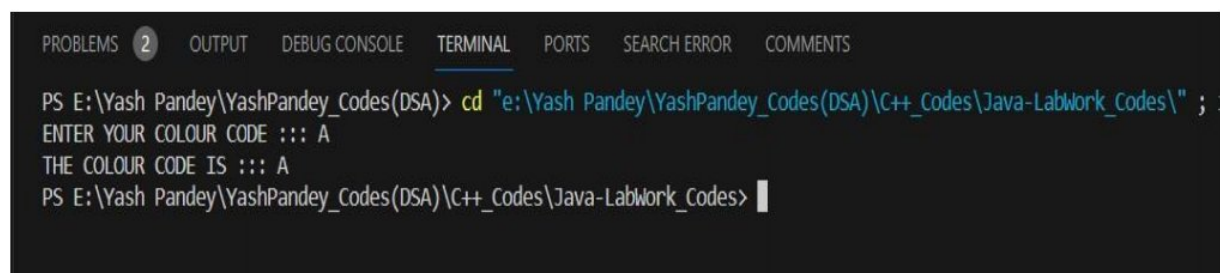
        char Char1 = sc.nextLine().charAt(0);

        System.out.println("THE COLOUR CODE IS ::: " + Char1);

    }

}
```

OUTPUT : -



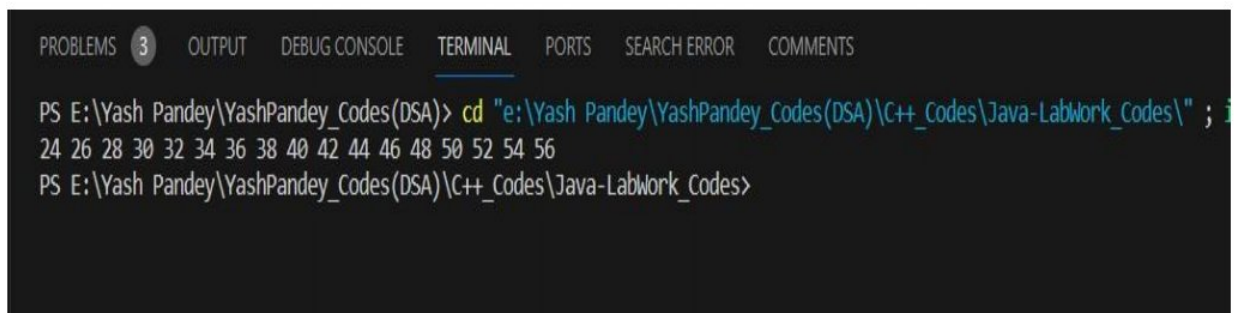
```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\" ;
ENTER YOUR COLOUR CODE ::: A
THE COLOUR CODE IS ::: A
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes> |
```

Program – 07.

Q :- WAP To Print Even N.o B/w 23 And 57 ?

```
public class Prog07 {  
    public static void main(String args[]) {  
        int a = 23, b = 57;  
        for (int i = a; i <= b; i++) {  
            if (i % 2 == 0) {        System.out.print(i + " ");        }  
        }  
    }  
}
```

OUTPUT : -



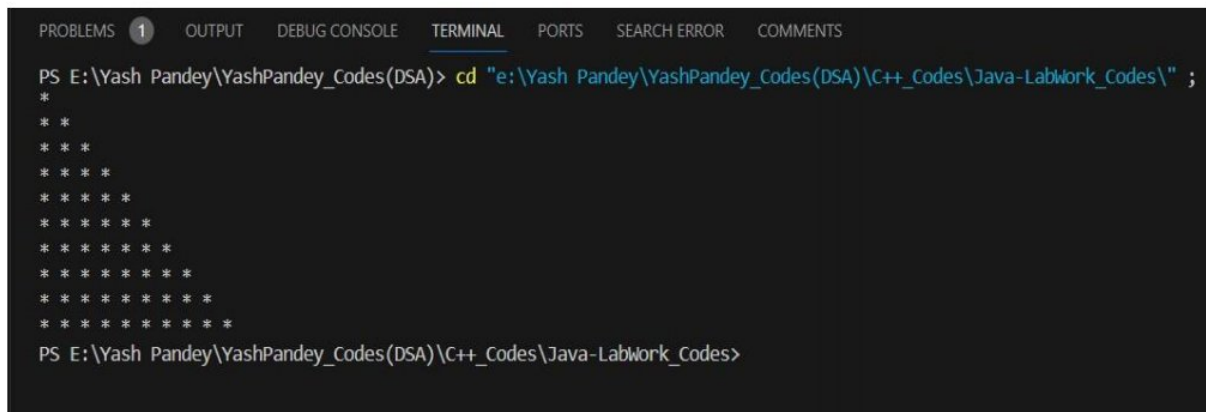
```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ; java Prog07  
24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 08.

Q :- WAP To Display * In Floyd's Pattern ?

```
public class Prog08 {  
    public static void main(String args[]){  
        for(int i=0 ; i<10 ; i++){  
            for(int j=0 ; j<=i ; j++){  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

OUTPUT :-



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\" ;  
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * * *  
* * * * * * *  
* * * * * * * *  
* * * * * * * * *  
* * * * * * * * * *  
* * * * * * * * * * *  
* * * * * * * * * * * *  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes>
```


Program – 09.

Q :- WAP To Check String Is Palindrome Or Not?

```
import java.util.*;

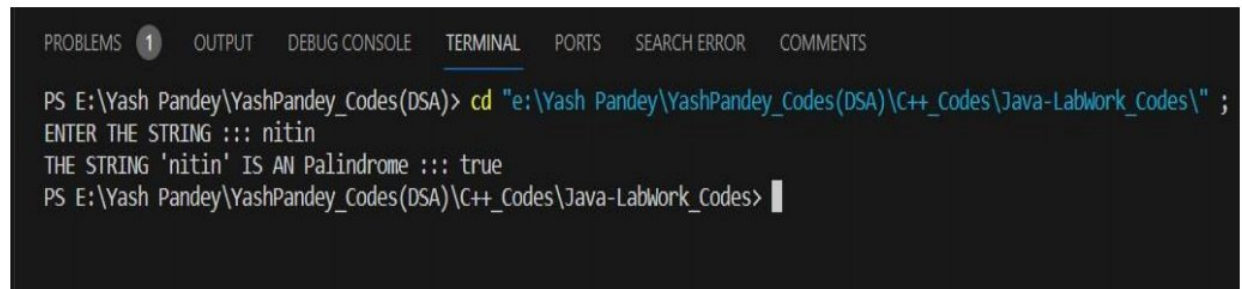
public class Prog09 {

    public static boolean Palindrome(String str) {
        int Start = 0, End = (str.length() - 1);
        while (Start <= End) {
            if (str.charAt(Start) != str.charAt(End)) {
                return false;
            }
            Start++;    End--;
        }
        return true;
    }

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("ENTER THE STRING ::: ");
```

```
String string = sc.nextLine();  
boolean ans = Palindrome(string);  
System.out.println("THE STRING '" + string + "'  
                    IS AN Palindrome ::: " + ans);  
}  
}
```

OUTPUT : -



The screenshot shows a code editor interface with a dark theme. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', 'PORTS', 'SEARCH ERROR', and 'COMMENTS'. The 'TERMINAL' tab is active. The terminal output shows the following sequence of commands and results:

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
ENTER THE STRING ::: nitin  
THE STRING 'nitin' IS AN Palindrome ::: true  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes> |
```

Program – 10.

Q :- WAP To Display ASCII Values Of Element Of Given Integer Array ?

```
public class Prog10 {  
    public static void main(String args[]){  
        int arr[]={67,73,84,92,97,99,104,109,116,122};  
        for(int i=0;i<arr.length;i++){  
            System.out.println("Char Value Of ASCII-"+arr[i]+"  
                                IS ::: "+(char)arr[i]);  
        }  
    }  
}
```

OUTPUT : -



```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
Prog10 }  
Char Value Of ASCII-67 IS ::: C  
Char Value Of ASCII-73 IS ::: I  
Char Value Of ASCII-84 IS ::: T  
Char Value Of ASCII-92 IS ::: \  
Char Value Of ASCII-97 IS ::: a  
Char Value Of ASCII-99 IS ::: c  
Char Value Of ASCII-104 IS ::: h  
Char Value Of ASCII-109 IS ::: m  
Char Value Of ASCII-116 IS ::: t  
Char Value Of ASCII-122 IS ::: z  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 11.

Q :- WAP To Reverse A Given 2X2 Matrix ?

```
import java.util.*;

public class Prog11 {

    public static void Reverse_Matrix(int[][] Matrix) {
        int Rows = (Matrix.length - 1), Cols = (Matrix[0].length-1);
        int ptr1 = 0, ptr2 = Rows;
        while (ptr1 <= ptr2) {
            for (int i = 0; i <= Cols; i++) {
                int temp = Matrix[ptr1][i];
                Matrix[ptr1][i] = Matrix[ptr2][Cols - i];
                Matrix[ptr2][Cols - i] = temp;
            }
            ptr1++;
            ptr2--;
        }
    }
}
```

```

public static void Display(int[][] Matrix) {
    for (int i = 0; i < Matrix.length; i++) {
        for (int j = 0; j < Matrix[0].length; j++) {
            System.out.print(Matrix[i][j] + " ");
        }
        System.out.println();
    }
}

public static void main(String args[]) {
    int[][] Matrix = { { 1, 2 }, { 4, 5 } };
    System.out.println("Before Swap , The 2X2 Matrix IS :::");
    Display(Matrix);      Reverse_Matrix(Matrix);
    System.out.println("After Swap , The 2X2 Matrix IS ::: ");
    Display(Matrix);
}
}

```

OUTPUT : -

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
Before Swap , The 2X2 Matrix IS :::
1 2
4 5

After Swap , The 2X2 Matrix IS :::
5 4
2 1
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>

```


Program – 12.

Q :- WAP To Implement Stack Using LinkedList ?

```
class Node {  
    int data;    Node next;  
    public Node(int d) {  
        data = d;        next = null;  
    }  
}  
  
public class Stack_LL {  
    static class Stacks {  
        public static Node Top;  
        public static int Size;  
        boolean isEmpty() {        return (Top == null);        }  
        void Push(int data) {  
            Node NewNode = new Node(data);  
            if (isEmpty()) {  
                Top = NewNode;        Size++;  
                return;  
            }  
        }  
    }  
}
```

```

        NewNode.next = Top;        Top = NewNode;

        Size++;
    }

    void Pop() {
        if (IsEmpty()) {
            System.out.println("THE STACK IS EMPTY ,
                                DELETION NOT POSSIBLE");

            return;
        }

        Node Temp = Top;    Top = Top.next;

        Temp.next = null;    Size--;
    }

    void Peek() {
        if (IsEmpty()) {
            System.out.println("THE STACK IS EMPTY ,
                                PEEK NOT POSSIBLE");

            return;
        }

        System.out.println("THE TOP OF THE IS ::: " + Top.data);
    }

    void Size_Stack() {
        System.out.println("THE SIZE OF THE STACK IS ::: " + Size);
    }

```

```

    void Display() {
        System.out.print("THE STACK IS ::: (Top) --> ");
        Node temp = Top;
        while (temp != null) {
            System.out.print(temp.data + " ");
            temp = temp.next;
        }
        System.out.println();
    }
}

public static void main(String args[]) {
    Stacks S1 = new Stacks();

    S1.Push(1);    S1.Push(2);    S1.Push(3);
    S1.Push(4);    S1.Push(5);
    S1.Display();  S1.Pop();       S1.Display();
    S1.Peek();     S1.Size_Stack();

}
}

```

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\
THE STACK IS ::: (Top) --> 5 4 3 2 1
THE STACK IS ::: (Top) --> 4 3 2 1
THE TOP OF THE IS ::: 4
THE SIZE OF THE STACK IS ::: 4

PS E:\Yash Pandey\YashPandey_Codes(DSA)\Java_Codes\Java1_BasisCodes>

```

Program – 13.

Q :- WAP To Implement Queue Using LinkedList ?

```
class Node {  
    int data;    Node next;  
    public Node(int d) {  
        data = d;        next = null;  
    }  
}  
  
public class Queues_LL {  
    static class Queues {  
        public static Node Front;  
        public static Node Rear;  
        public static int Size;  
        boolean isEmpty() {        return (Rear == null);        }  
        void Enqueue(int data) {  
            Node NewNode = new Node(data);  
            if (isEmpty()) {  
                Front = Rear = NewNode;  
                Size++;    return;  
            }  
        }  
    }  
}
```

```

        Rear.next = NewNode;        Rear = NewNode;
        Size++;
    }

    void Dequeue() {
        if (IsEmpty()) {
            System.out.println("THE STACK IS EMPTY ,
                                DELETION NOT POSSIBLE");

            return;
        }

        Node Temp = Front;        Front = Front.next;
        Temp.next = null;        Size--;
    }

    void Peek_First() {
        if (IsEmpty()) {
            System.out.println("THE QUEUE IS EMPTY ,
                                PEEK NOT POSSIBLE");

            return;
        }

        System.out.println("THE FRONT OF THE IS ::: " + Front.data);
    }

    void Peek_Last() {
        if (IsEmpty()) {
            System.out.println("THE QUEUE IS EMPTY ,
                                PEEK NOT POSSIBLE");

```

```

        return;
    }
    System.out.println("THE REAR OF THE IS ::: " + Rear.data);
}
void Size_Queue() {
    System.out.println("THE SIZE OF THE QUEUE IS ::: " + Size);
}
void Display() {
    System.out.print("THE QUEUE IS ::: (Front) --> ");
    Node temp = Front;
    while (temp != null) {
        System.out.print(temp.data + " ");
        temp = temp.next;
    }
    System.out.println("<-- (Rear)");
}
}

public static void main(String args[]) {
    Queues q = new Queues();
    q.Enqueue(1);
    q.Enqueue(2);
    q.Enqueue(3);
    q.Enqueue(4);
    q.Enqueue(5);
}

```



```
        q.Display();  
        q.Dequeue();  
        q.Display();  
        q.Peek_First();  
        q.Peek_Last();  
        q.Size_Queue();  
    }  
}
```

OUTPUT :

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\Java_Codes\Java1_BasisCodes\" ;  
  
THE QUEUE IS ::: (Front) --> 1 2 3 4 5 <-- (Rear)  
THE QUEUE IS ::: (Front) --> 2 3 4 5 <-- (Rear)  
THE FRONT OF THE QUEUE IS ::: 2  
THE REAR OF THE QUEUE IS ::: 5  
THE SIZE OF THE QUEUE IS ::: 4  
  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\Java_Codes\Java1_BasisCodes>
```

Program – 14.

Q :- WAP To produce tokens from a given long string?

```
import java.util.*;

public class Prog14 {

    String input;

    Prog14() {

        Scanner reader = new Scanner(System.in);

        System.out.println("Enter your desired string");

        String input = reader.nextLine();

    }

    Prog14(String input) {    this.input = input;    }

    int count() {

        int token = 1;

        for (int i = 0; i < input.length(); i++) {

            if (input.charAt(i) == ' ') {    token++;    }

        }    return token;

    }

    String[] tokenize() {

        String tokens[] = new String[count()];

        int prev = 0, k = 0;

        for (int i = 0; i < input.length(); i++) {
```

```

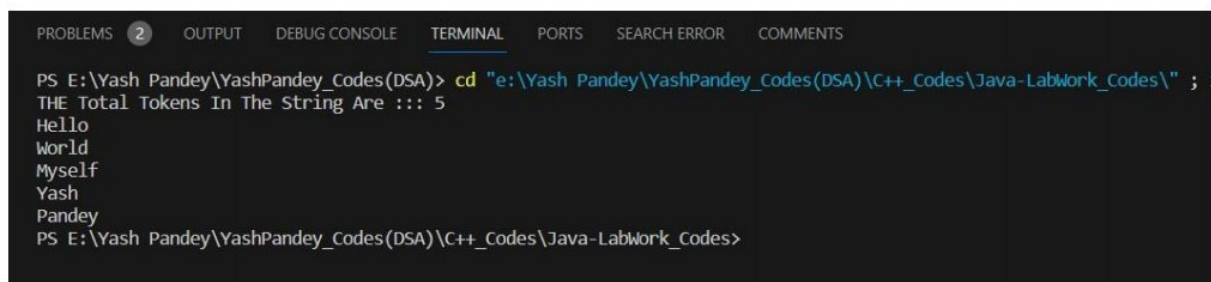
        if (input.charAt(i) == ' ') {
            tokens[k++] = input.substring(prev, i);
            prev = i + 1;
        }
    }
    tokens[k] = input.substring(prev, input.length());
    return tokens;
}

public static void main(String[] args) {
    Prog14 t1= new Prog14("Hello World Myself Yash Pandey");
    System.out.println("THE Total Tokens In The String Are ::: "
        + t1.count());

    String tokens[] = t1.tokenize();
    for (int i = 0; i < tokens.length; i++) {
        System.out.println(tokens[i]);
    }
}
}

```

OUTPUT : -



```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
THE Total Tokens In The String Are ::: 5
Hello
World
Myself
Yash
Pandey
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>

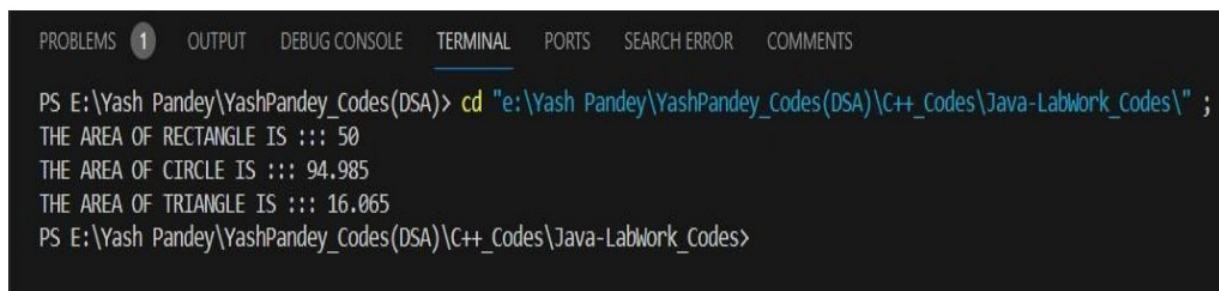
```

Program – 15.

Q :- WAP To Implement Method Overloading In Java ?

```
public class Prog15 {  
    public static void Area(int l, int b) {  
        System.out.println("AREA OF Rectangle IS ::: "+(l*b));  
    }  
    public static void Area(double r) {  
        System.out.println("AREA OF CIRCLE IS ::: "+(3.14*r*r));  
    }  
    public static void Area(double b, double h) {  
        System.out.println("AREA OF Triangle IS:::"+(0.5*b*h));  
    }  
    public static void main(String args[]) {  
        Area(5, 10);        Area(5.5);        Area(5.1, 6.3);  
    }  
}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
THE AREA OF RECTANGLE IS ::: 50  
THE AREA OF CIRCLE IS ::: 94.985  
THE AREA OF TRIANGLE IS ::: 16.065  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 16.

Q :- WAP To implement ATM transaction to dispense Notes of various denominations to the user.?

```
import java.util.*;

public class Prog16 {
    int amount;
    int[] denom;

    Prog16() {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the amount:");
        amount = input.nextInt();
        System.out.print("Enter the number of denominations:");
        denom = new int[input.nextInt()];
        System.out.println("Enter the denomination");
        for (int i = 0; i < denom.length; i++) {
            denom[i] = input.nextInt();
        }
        Arrays.sort(denom);
    }
}
```

```

int count() {
    int[] output = notes();
    int sum = 0 , end= output.length;
    for (int i = 0; i < end; i++) {    sum += output[i];    }
    return sum;
}

int[] notes() {
    int[] result = new int[denom.length] , i=0;
    while (i < denom.length - 1) {
        int j = 1;
        while ((amount - denom[i] * j) % denom[i + 1] != 0) {
            j++;
        }
        result[i] = j;
        amount -= denom[i] * j;        i++;
    }

    // Assuming amount is divisible by the last denom
    result[i] = amount / denom[i];
    return result;
}

```



```
Prog16(int amount) {  
    this.amount = amount;  
    this.denom = new int[] { 10, 20, 50, 100, 500 };  
}  
  
public static void main(String[] args) {  
    Prog16 t1 = new Prog16();  
    System.out.println(Arrays.toString(t1.notes()));  
}  
}
```

OUTPUT : -

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
Enter the amount:3850  
Enter the number of denominations:3  
Enter the denomination  
50  
100  
500  
[1, 3, 7]  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes> |
```

Program – 17.

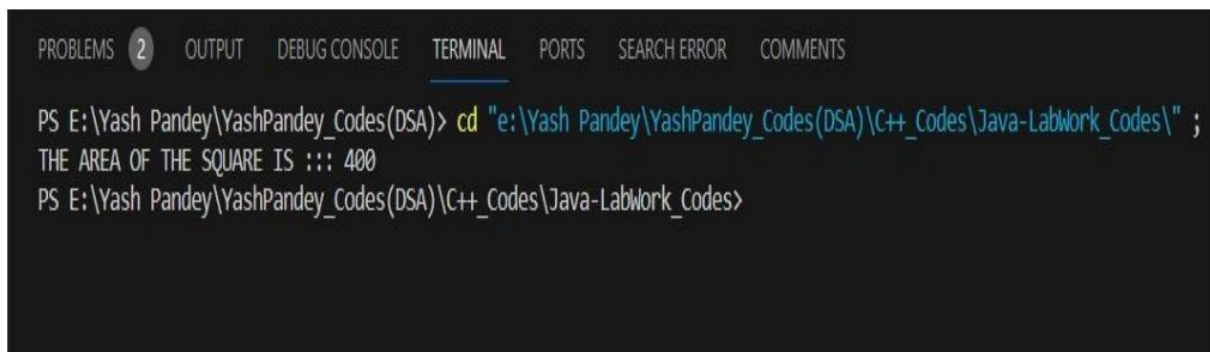
Q :- WAP to create an class Box uses parameterized constructor to initialize the dimensions of a box. The dimensions of the Box are width, height, depth. The Class should have method to return volume.

```
import java.util.*;

class Box {
    int Height , Width , Depth;
    public Box(int h, int w, int d) {
        this.Height = h;
        this.Width = w;
        this.Depth = d;
    }
    int Volume() {
        return (Height * Width * Depth);
    }
}
```

```
public class Prog17 {  
    public static void main(String args[]) {  
        Box Square = new Box(5, 8, 10);  
        int ans = Square.Volume();  
        System.out.println("THE AREA OF THE SQUARE IS  
        ::: " + ans);  
    }  
}
```

OUTPUT : -



```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
THE AREA OF THE SQUARE IS ::: 400  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

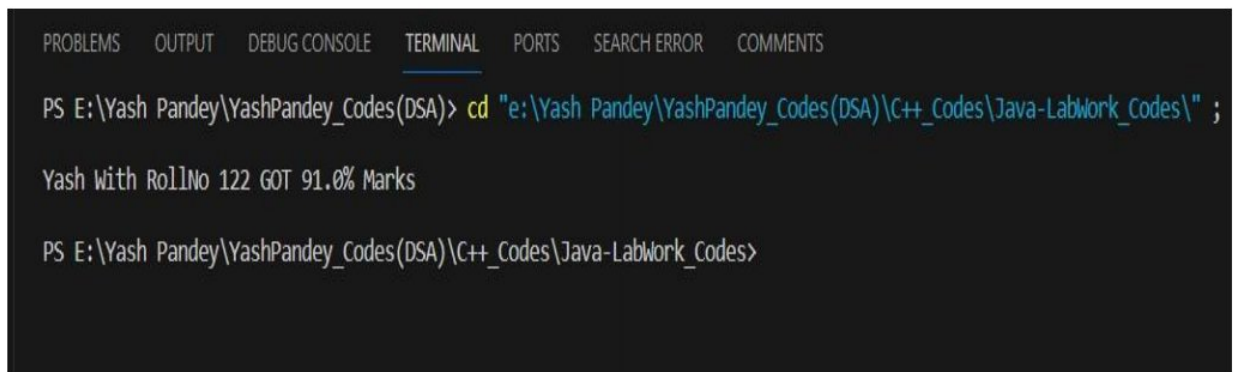
Program – 18.

Q :- WAP To Display The Use Of 'This' Keyword ?

```
class Student {  
    int RollNo;  
    String Name;  
    float Percentage;  
    public Student(int r, String n, float p) {  
        this.RollNo = r;  
        this.Name = n;  
        this.Percentage = p;  
    }  
    void Result() {  
        System.out.println(Name + " With RollNo " +  
            RollNo + " GOT " + Percentage + "% Marks");  
    }  
}
```

```
public class Prog18 {  
    public static void main(String args[]) {  
        Student Yash = new Student(122, "Yash", 91);  
        Yash.Result();  
    }  
}
```

OUTPUT : -



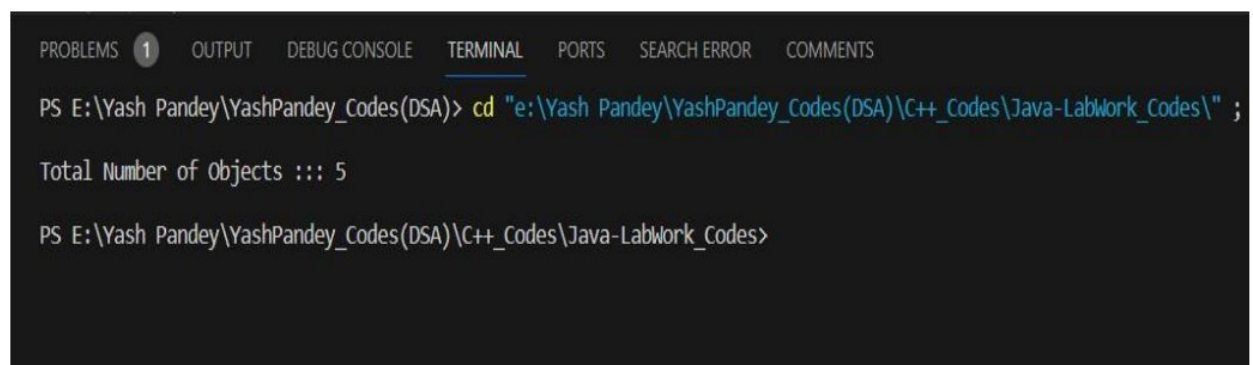
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH ERROR  COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\" ;  
Yash With RollNo 122 GOT 91.0% Marks  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes>
```

Program – 19.

Q :- WAP To Count The No. Of Instances Created For An Class ?

```
class Numbers {  
    int Num;        static int Count;  
    public Numbers() {        Count++;        }  
}  
  
public class Prog19 {  
    public static void main(String args[]) {  
        Numbers[] ob1 = new Numbers[5] ;  
        System.out.println("Total Number of Objects ::: " +  
                            Numbers.Count);  
    }  
}
```

OUTPUT : -



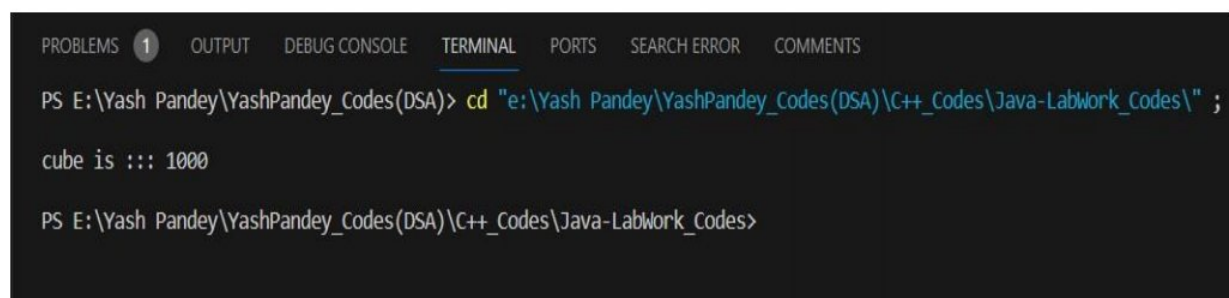
```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
Total Number of Objects ::: 5  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```


Program – 20.

Q :- WAP to Get Cube Of A Number By Static Method?

```
class cube {  
    static int num;  
    cube(int n) {        num = n;        }  
    static void Calculate() {  
        System.out.println("cube is ::: " + (num*num*num));  
    }  
}  
  
public class Prog20 {  
    public static void main(String args[]) {  
        cube c1 = new cube(10);  
        c1.Calculate();  
    }  
}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
cube is ::: 1000  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 21.

Q :- WAP To Implement Method Overriding ?

```
class Base {  
    void display() {  
        System.out.println("this is Base class Function");  
    }  
}  
class Derived extends Base {  
    void display() {  
        System.out.println("this is Derived class Function");  
    }  
}  
  
public class Prog21 {  
    public static void main(String args[]) {  
        Derived c1 = new Derived();        c1.display();  
    }  
}
```

OUTPUT :-

A screenshot of a code editor interface. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', 'PORTS', 'SEARCH ERROR', and 'COMMENTS'. The 'TERMINAL' tab is active. The terminal shows the command 'cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;' followed by the output 'this is Derived class Function'. The prompt 'PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>' is visible at the bottom.

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
  
this is Derived class Function  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

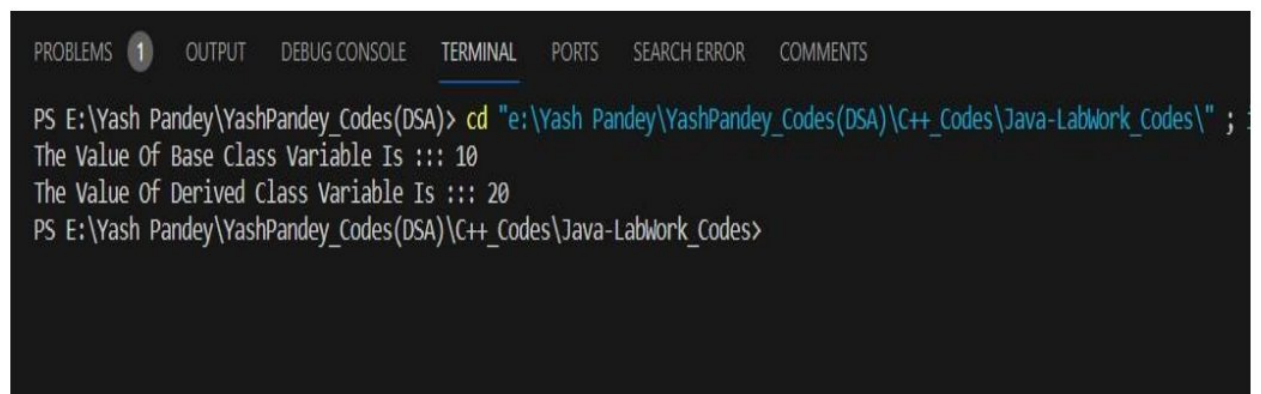
Program – 22.

Q :- WAP To Implement Single Inheritance ?

```
public class Prog22 {  
    static class Base {  
        int i;  
  
        public Base() {            i = 0;            }  
        public Base(int a) {        i = a;            }  
    }  
  
    static class Derived extends Base {  
        int j;  
  
        public Derived() {  
            super();    j = 0;  
        }  
  
        public Derived(int a, int b) {  
            super(a);j = b;  
        }  
    }  
}  
  
public static void main(String[] args) {  
    Derived d = new Derived(10, 20);  
    System.out.println("The Value Of Base Class  
                        Variable Is ::: " + d.i);  
}
```

```
        System.out.println("The Value Of Derived Class  
Variable Is ::: " + d.j);  
    }  
}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\" ; g++ 1.cpp && .\1.exe  
The Value Of Base Class Variable Is ::: 10  
The Value Of Derived Class Variable Is ::: 20  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes>
```

Program – 23.

Q :- WAP To Implement MultiLevel Inheritance ?

```
public class Prog23 {  
    static class Base {  
        int i;  
        public Base() {      i = 0;      }  
        public Base(int a) {    i = a;    }  
    }  
    static class Intermediate extends Base {  
        int j;  
        public Intermediate() {  
            super();      j = 0;  
        }  
        public Intermediate(int a, int b) {  
            super(a);      j = b;  
        }  
    }  
    static class Derived extends Intermediate {  
        int k;
```

```

        public Intermediate() {
            super();          k = 0;
        }

        public Intermediate(int a, int b, int c) {
            super(a,b);       k = c;
        }
    }

    public static void main(String[] args) {
        Derived d = new Derived(10, 20, 30);
        System.out.println("The Value Of Base Class
                           Variable Is ::: " + d.i);
        System.out.println("The Value Of Intermediate
                           Class Variable Is ::: " + d.j);
        System.out.println("The Value Of Derived Class
                           Variable Is ::: " + d.k);
    }
}

```

OUTPUT : -

```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
The Value Of Base Class Variable Is ::: 10
The Value Of Intermediate Class Variable Is ::: 20
The Value Of Derived Class Variable Is ::: 30
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>

```

Program – 24.

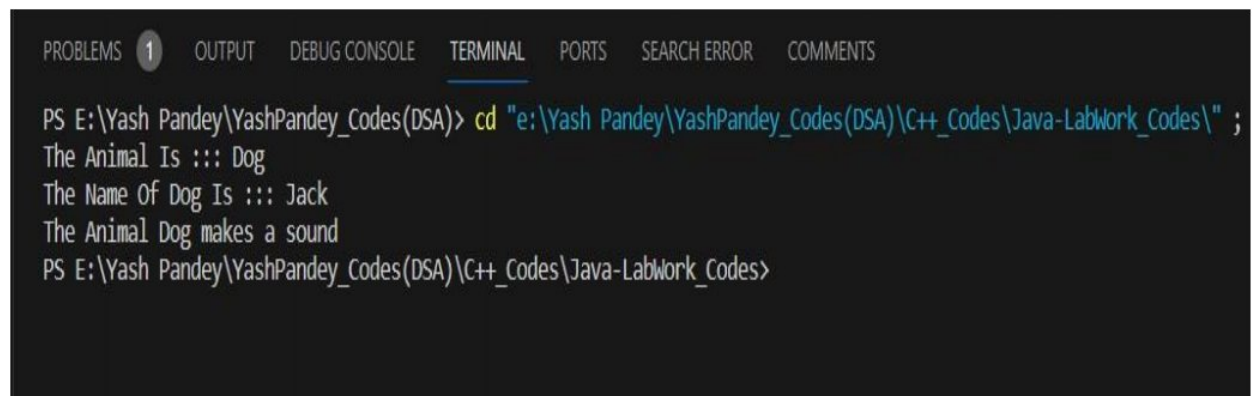
Q :- WAP To Implement Super() Method In Java ?

```
class Animal {  
    String Animal;  
    public Animal(String a) {  
        this.Animal = a;  
        System.out.println("The Animal Is ::: " + Animal);  
    }  
    public void AnimalSound() {  
        System.out.println("The Animal " + Animal + "  
                            makes a sound");  
    }  
}  
  
class Dog extends Animal {  
    String Dog;  
    public Dog(String a, String d) {  
        super(a);  
        this.Dog = d;  
        System.out.println("The Name Of Dog Is ::: " + Dog);  
    }  
}
```



```
        public void AnimalSound() {  
            super.AnimalSound();  
        }  
    }  
  
    public class Prog24 {  
        public static void main(String args[]) {  
            Animal myDog = new Dog("Dog", "Jack");  
            myDog.AnimalSound();  
        }  
    }  
}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
The Animal Is ::: Dog  
The Name Of Dog Is ::: Jack  
The Animal Dog makes a sound  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

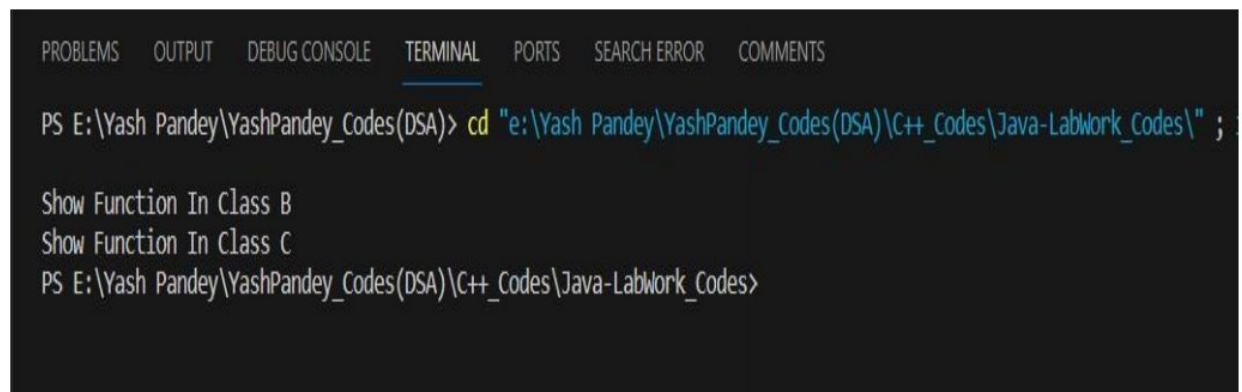
Program – 25.

Q:- WAP To Show Dynamic Polymorphism & Interface?

```
interface A {  
    // Interface definition For Abstract method  
    void Show();  
}  
  
class B implements A {  
    @Override  
    public void Show() {  
        System.out.println("Show Function In Class B");  
    }  
}  
  
class C implements A {  
    @Override  
    public void Show() {  
        System.out.println("Show Function In Class C");  
    }  
}
```

```
public class Prog25 {  
    public static void main(String[] args) {  
        A Obj = new B();  
        Obj.Show();  
        Obj = new C();  
        Obj.Show();  
    }  
}
```

OUTPUT : -



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH ERROR  COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
  
Show Function In Class B  
Show Function In Class C  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 26.

Q :- WAP To Create An Abstract Class Shape Where Rectangle And Triangle Inherit The Shape Class ?

```
abstract class Shape {  
    int Base, Height;  
    public abstract void Show();  
    public abstract void Area();  
}  
  
class Rectangle extends Shape {  
    public Rectangle(int b, int h) {  
        Base = b;           Height = h;  
    }  
    @Override  
    public void Show() {  
        System.out.println("The Rectangle Has Base=" +  
            Base + " And Height=" + Height);  
    }  
    public void Area() {  
        System.out.println("The Area Of Rectangle Is :::  
            " + (Base * Height));    }  
}
```

```
class Triangle extends Shape {  
    public Triangle (int b, int h) {  
        Base = b;           Height = h;  
    }  
    @Override  
    public void Show() {  
        System.out.println("The Triangle Has Base=" +  
            Base + " And Height=" + Height);  
    }  
    public void Area() {  
        System.out.println("The Area Of Triangle Is ::  
            " + (0.5*Base* Height));  
    }  
}
```

```
public class Prog26 {  
    public static void main(String args[]) {  
        Shape s = new Rectangle(10, 20);  
        s.Show();  
        s.Area();  
    }  
}
```

```
s = new Triangle(20, 20);  
s.Show();  
s.Area();  
}  
}
```

OUTPUT : -



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
  
The Rectangle Has Base=10 And Height=20  
The Arae Of Rectangle Is ::: 200  
  
The Triangle Has Base=20 And Height=20  
The Arae Of Triangle Is ::: 200.0  
  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 27.

Q:- Write An Java Package Show Dynamic Polymorphism & Interface In Java ?

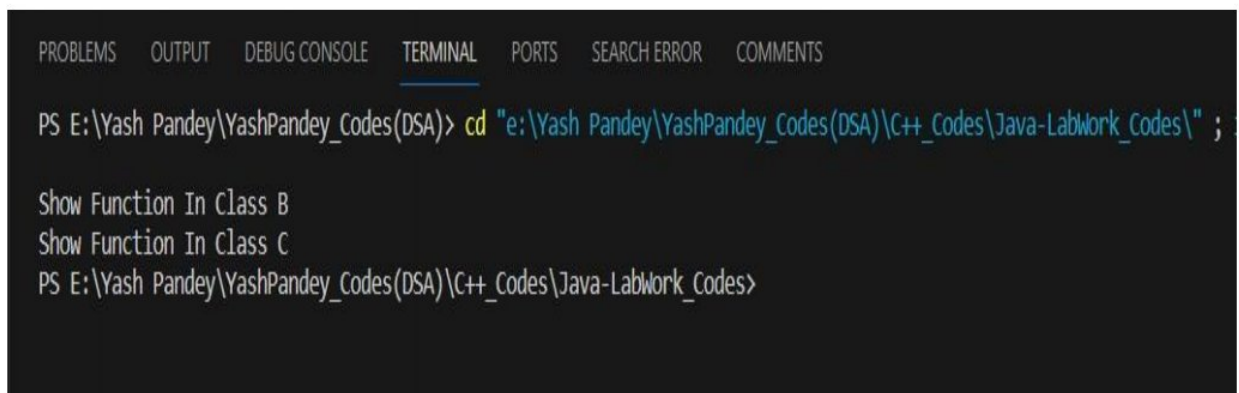
Package Polymorphism ;

```
interface A {  
    void Show();  
}  
  
class B implements A {  
    @Override  
    public void Show() {  
        System.out.println("Show Function In Class B");  
    }  
}  
  
class C implements A {  
    @Override  
    public void Show() {  
        System.out.println("Show Function In Class C");  
    }  
}
```



```
public class Prog27 {  
    public static void main(String[] args) {  
        A Obj = new B();  
        Obj.Show();  
        Obj = new C();  
        Obj.Show();  
    }  
}
```

OUTPUT : -



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH ERROR  COMMENTS  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;  
  
Show Function In Class B  
Show Function In Class C  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

Program – 28.

Q :- Write An Application That Creates An Interface And Implements It ?

```
interface Shape { // Interface Shape....
    void Area();
}

class Circle implements Shape {
    double Radius;
    public Circle(double r) {    Radius = r;    }
    @Override
    public void Area() {
        System.out.println("The Area Of Circle Is :: " +
                            (3.14 * Radius * Radius));
    }
}

public class Prog28 {
    public static void main(String[] args) {
        Shape s = new Circle(5);
        s.Area();
    }
}
```

OUTPUT : -

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
```

```
The Area Of Circle Is ::: 78.5
```

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```



Program – 29.

Q:- Write an **interface playable** with a method **void play()**. let this Interface be placed in a package called **music** , write a **class Veena** which implement the playable interface. Let this class be placed in a package called **music.string** . Write a **class saxophone** which implement the playable interface, let this class be placed in a package called **music.wind** . Write another **class test** in package **live** . Then :-

- (A) create an instance of Veena and call the play() method.
- (B) create an instance of saxophone and Call the play() method.
- (C) place the above instances in a variable of type playable and then call play().?

Hierarchy Of Packages



1. Package Music

➤ Playable.java File

```
J Playable.java U ×
Music > J Playable.java > ...
Click here to ask Blackbox to help you code faster |
1 package Music;
2
3 public interface Playable {
4     void Play();
5 }
6
```

➤ Package String Having Veena.java File

```
J Veena.java U ●
Music > String > J Veena.java > ...
Click here to ask Blackbox to help you code faster |
1 package Music.String;
2 import Music.Playable;
3
4 public class Veena implements Playable {
5     String Name;
6     public Veena(String n) {
7         this.Name = n;
8     }
9     @Override
10    public void Play() {
11        System.out.println(Name + " Can Play Veena Well !");
12    }
13    public static void main(String args[]) {
14        Playable p = new Veena("Yash");
15        p.Play();
16    }
17 }
18
```

➤ Package Wind Having Saxophone.java File

```
J Saxophone.java U •
Music > Wind > J Saxophone.java > ...
💡 Click here to ask Blackbox to help you code faster |

1 package Music.Wind;
2 import Music.Playable;
3
4 public class Saxophone implements Playable {
5     String Name;
6     public Saxophone(String n) {
7         this.Name = n;
8     }
9     @Override
10    public void Play() {
11        System.out.println(Name + " Can Play Saxophone Well !");
12    }
13    public static void main(String args[]) {
14        Playable p = new Saxophone("Yash");
15        p.Play();
16    }
17 }
18
```

2. Package Live Having Test.java File

```
J Test.java U •
Live > J Test.java > ...
💡 Click here to ask Blackbox to help you code faster |

1 package Live;
2 import Music.Playable;
3 import Music.String.Veena;
4 import Music.Wind.Saxophone;
5
6 public class Test {
7     public static void main(String args[]) {
8         System.out.print("Instance Of Veena Class ::: ");
9         Veena v = new Veena("Yash");
10        v.Play();
11        System.out.print("Instance Of Saxophone Class ::: ");
12        Saxophone s = new Saxophone("Yash");
13        s.Play();
14        System.out.print("Instance Of Playable Interface ::: ");
15        Playable p = new Veena("Yash");
16        p.Play();
17        System.out.print("Instance Of Playable Interface ::: ");
18        p = new Saxophone("Yash");
19        p.Play();
20    }
21 }
22
```


OUTPUT : -

```
Instance Of Veena Class ::: Yash Can Play Veena Well !  
Instance Of Saxophone Class ::: Yash Can Play Saxophone Well !  
  
Instance Of Playable Interface ::: Yash Can Play Veena Well !  
Instance Of Playable Interface ::: Yash Can Play Saxophone Well !
```


Program – 30.

Q :- WAP to accept name and age of a person from the user . Ensure that entered age $\geq$ 15 & age $<$ 60 . Display Proper Error message And The program must execute gracefully after displaying the error message in case the argument pass is not proper ?

```
import java.util.*;

class MyException extends Exception {
    public MyException(String message) {
        super(message);
    }
}

public class Prog30 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        String name;    int Age;
        try {
            System.out.print("Enter your Name ::: ");
            name = sc.nextLine();
            System.out.print("Enter your Age ::: ");
            Age = sc.nextInt();
```

```
        if (Age < 15) {
            throw new MyException(name + " Age Is "
                + Age + " Which Is Less Than 15 Years");
        }
        else if (Age > 60) {
            throw new MyException(name + " Age Is "
                + Age + " Which Is More Than 60 Years");
        } else {
            System.out.print(name+" Age Is ::: "+Age);
        }
    }
    catch (Exception e) {
        System.out.println("Error : " + e);
    }
    finally {
        System.out.println("Program Exicuted.....");
    }
}
```

OUTPUT :-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH ERROR  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\"

Enter your Name ::: Yash
Enter your Age ::: 12

Error : MyException: Yash Age Is 12 Which Is Less Than 15 Years
Program Exicuted.....

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C

Enter your Name ::: Aman
Enter your Age ::: 21

Aman Age Is ::: 21
Program Exicuted.....

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C

Enter your Name ::: John
Enter your Age ::: 64

Error : MyException: John Age Is 64 Which Is More Than 60 Years
Program Exicuted.....

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes> █
```

Program – 31.

Q :- Write To Create An Customized Exception & Also Make Use Of All The 5 Exception Keywords ?

```
Import java.util.*;

class MyException extends Exception {
    public MyException(String message) {
        super(message);
    }
}

public class Prog31 {
    public static void CheckException(String Name, int Age) {
        try {
            Check(Name, Age);
        }
        catch (MyException e) {
            System.out.println("Error : " + e);
        }
        finally {
            System.out.println("Program Exicuted.....");
        }
    }
}
```

```

    public void Check(String Name, int Age) throws
    MyException {
        if (Age < 18) {
            throw new MyException(Name + " Is Not
                                   Eligible For Voting");
        } else {
            System.out.println(Name+" Is Eligible For Voting");
        }
    }

    public static void main(String args[]) {
        CheckException("Yash", 21);
        CheckException("Rahul", 17);
    }
}

```

OUTPUT :-

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH ERROR  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;
Yash Is Eligible For Voting
Program Exicuted.....

Error : MyException: Rahul Is Not Eligible For Voting
Program Exicuted.....

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>

```

Program – 32.

Q :- Write An Applet Program That Displays “Hello World” With Background Color As Black , text Color Blue And Your Name In The Status Window ?

```
import java.applet.Applet;
import java.awt.*;

public class Prog32 extends Applet {
    public void paint(Graphics g) {
        g.setColor(Color.blue);
        g.drawString("Hello World", 50, 50);
    }
}
```

‘Prog32.html’ File :-

A screenshot of a code editor window titled 'Prog32.html U x'. The editor shows the following HTML code:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Yash Prog-32</title>
  </head>
  <body>
    <applet code="Prog32.class" width="300" height="300">
    </applet>
  </body>
</html>
```

The code is displayed with line numbers 1 through 12 on the left margin. The editor interface includes a breadcrumb path 'C++_Codes > Java-LabWork_Codes > Prog32.html' and a help prompt 'Click here to ask Blackbox to help you code faster'.

OUTPUT :-

Hello World

Program – 33.

Q :- Develop An Analog Clock Using Applet ?

```
import java.applet.Applet;
import java.awt.*;
import java.util.*;

public class Prog33 extends Applet {
    @Override
    public void init() {
        this.setSize(new Dimension(800, 400));
        setBackground(new Color(50, 50, 50));
        new Thread() {
            @Override
            public void run() {
                while (true) {
                    repaint();
                    delayAnimation();
                }
            }
        }.start();
    }
}
```

```

private void delayAnimation() {
    try {        Thread.sleep(1000);        }
    catch (Exception e) {            e.printStackTrace();            }
}

public void paint(Graphics g) {
    // Get the system time
    Calendar time = Calendar.getInstance();
    int hour = time.get(Calendar.HOUR_OF_DAY);
    int minute = time.get(Calendar.MINUTE);
    int second = time.get(Calendar.SECOND);
    if (hour > 12) {        hour = hour-12;        }
    // Draw clock body center at (400, 200)
    g.setColor(Color.white);
    g.fillOval(300, 100, 200, 200);
    // Labeling Of Clock Area
    g.setColor(Color.black);
    g.drawString("12", 390, 120);
    g.drawString("9", 310, 200);
    g.drawString("6", 400, 290);
    g.drawString("3", 480, 200);
}

```

// Declaring variables to be used

double angle; int x, y;

// Second hand's angle in Radian

angle = Math.toRadians((15 - second) * 6);

// Position of the second hand with length 100 unit

x = (int) (Math.cos(angle) * 100);

y = (int) (Math.sin(angle) * 100);

// Red color second hand

g.setColor(Color.red);

g.drawLine(400, 200, 400 + x, 200 - y);

// Minute hand's angle in Radian

angle = Math.toRadians((15 - minute) * 6);

// Position of the minute hand with length 80 ..

x = (int) (Math.cos(angle) * 80);

y = (int) (Math.sin(angle) * 80);

// blue color Minute hand

g.setColor(Color.blue);

g.drawLine(400, 200, 400 + x, 200 - y);

// Hour hand's angle in Radian

angle = Math.toRadians((15 - (hour * 5)) * 6);

// Position of the hour hand with length 50 .

```
x = (int) (Math.cos(angle) * 50);
```

```
y = (int) (Math.sin(angle) * 50);
```

// Black color hour hand

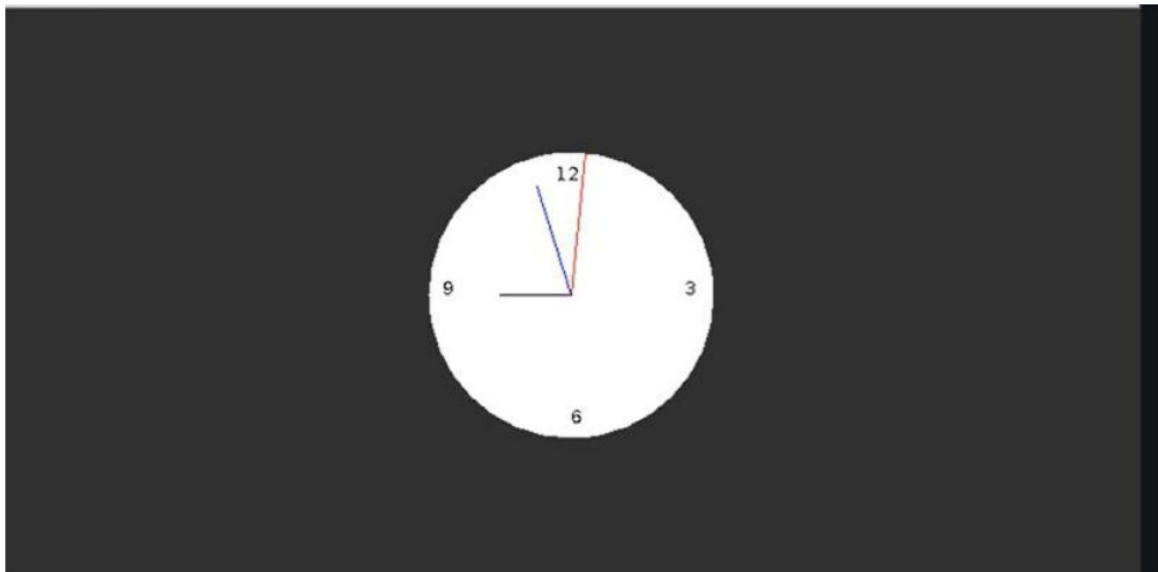
```
g.setColor(Color.black);
```

```
g.drawLine(400, 200, 400 + x, 200 - y);
```

```
}
```

```
}
```

OUTPUT : -



Program – 34.

Q :- Write An Java Program To Show Multi-Threaded Producer And Consumer Application ?

```
class Base {  
    int num;  
    boolean ValueSet = false;  
    public synchronized void Put(int n) {  
        while (ValueSet == true) {  
            try {        wait();        }  
            catch (Exception e) { }  
        }  
        System.out.println("Put --> num : " + n);  
        this.num = n;  
        ValueSet = true;  
        notify();  
    }  
    public synchronized void Get() {  
        while (ValueSet == false) {  
            try {        wait();        }  
            catch (Exception e) { }  
        }  
    }  
}
```

```

        System.out.println("Get --> num : " + num);
        ValueSet = false;          notify();
    }
}

class Producer implements Runnable {
    Base Obj;

    public Producer(Base b) {
        this.Obj = b;
        Thread T1 = new Thread(this, "Producer");
        T1.start();
    }

    @Override
    public void run() {
        int i = 0;
        while (i <= 5) {
            Obj.Put(i++);
            try {        Thread.sleep(1000);        }
            catch(Exception e){ e.printStackTrace(); }
        }
    }
}

```

class Consumer implements Runnable {

Base Obj;

public Consumer(Base b) {

 this.Obj = b;

 Thread T2 = new Thread(this, "Consumer");

 T2.start();

}

@Override

public void run() {

 int i = 0;

while (i <= 5) {

 Obj.Get();

try { Thread.sleep(2000); **}**

catch(Exception e){ e.printStackTrace(); **}**

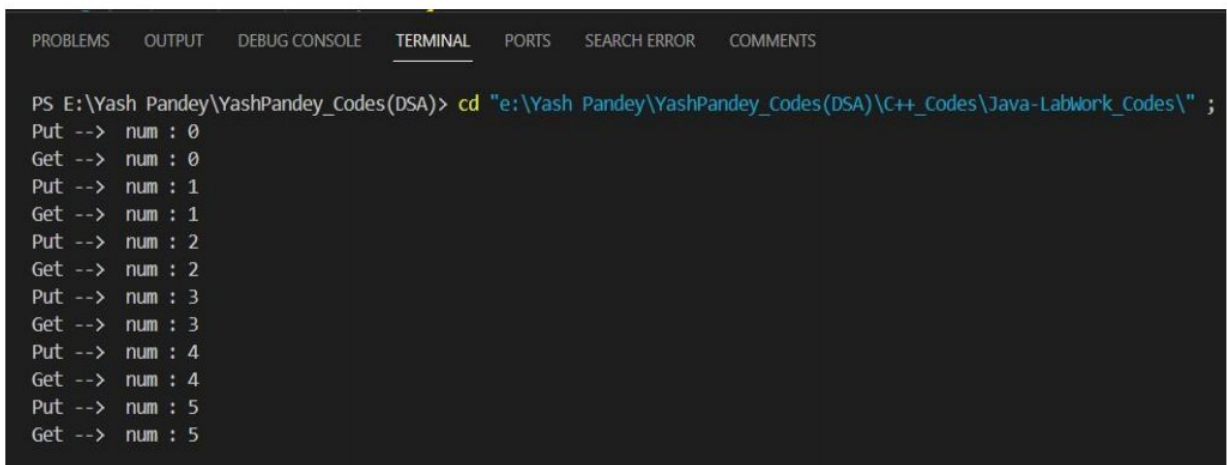
}

}

}


```
public class Prog34 {  
    public static void main(String args[]) {  
        Base Obj = new Base();  
        Producer P = new Producer(Obj);  
        Consumer C = new Consumer(Obj);  
    }  
}
```

OUTPUT : -



```
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-Labwork_Codes\" ;  
Put --> num : 0  
Get --> num : 0  
Put --> num : 1  
Get --> num : 1  
Put --> num : 2  
Get --> num : 2  
Put --> num : 3  
Get --> num : 3  
Put --> num : 4  
Get --> num : 4  
Put --> num : 5  
Get --> num : 5
```

Program – 35.

Q :- Write an application that executes to threads .
One Thread displays "An" After every 1000 ms and
the other displays "B" after every 3000 ms . Create the
Threads by executing the thread class .?

```
class First extends Thread{
    public void run(){
        for(int i=0 ; i<9 ; i++){
            System.out.print("An  ");
            try{ Thread.sleep(1000); }
            catch(Exception e){ }
        }
    }
}

class Second extends Thread{
    public void run(){
        for(int i=0;i<3;i++){
            System.out.println("\nB  ");
            try{ Thread.sleep(3000); }
            catch(Exception e){ }
        }
    }
}
```

```

    }
}

public class Prog35 {
    public static void main (String args[]){
        First F = new First();
        Second S = new Second();
        F.start();
        try{ Thread.sleep(3000); }
        catch(Exception e){ }
        S.start();
    }
}

```

OUTPUT :-



```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;

An   An   An
B
An   An   An
B
An   An   An
B
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>

```

Program – 36.

Q :- WAP And Create a class salesperson as a thread that will display an salesperson name . create a class days as other thread that has array of 7 days . call the instance of sales person in days and start both the threads . Suspend the sales person on Sunday and resume on Wednesday . We can only use suspend and resume methods from the thread only .?

```
class SalesPerson extends Thread {  
    public String name;  
    public SalesPerson(String n) {    this.name = n;    }  
    public void run() {  
        System.out.println("Salesperson " + name + "  
                             reporting for duty.");  
    }  
}  
  
class Days extends Thread {  
    String[] days = { "Sunday", "Monday", "Tuesday",  
                     "Wednesday", "Thursday", "Friday", "Saturday" };  
    SalesPerson salesPerson;
```

```
public Days(SalesPerson salesPerson) {  
    this.salesPerson = salesPerson;  
}  
public void run() {  
    for (String day : days) {  
        if (day.equals("Sunday")) {  
            System.out.println("Salesperson "+salesPerson.name  
                +" suspended Shop On : " + day);  
            salesPerson.suspend();  
        }  
        else if (day.equals("Wednesday")) {  
            System.out.println("Salesperson "+SalesPerson.name  
                + " resumed Shop On : " + day);  
            salesPerson.resume();  
        }  
        else {  
            System.out.println("Salesperson "+SalesPerson.name  
                + " Continued Shop On : " + day);  
        }  
    }  
}  
}
```

```

public class Prog36 {
    public static void main(String args[]) {
        SalesPerson SP = new SalesPerson("Yash");
        Days DP = new Days(SP);
        SP.start();
        DP.start();
        try{        SP.join();        DP.join();        }
        catch(Exception e){ }
    }
}

```

OUTPUT : -

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH ERROR  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;

Salesperson Yash reporting for duty.

Salesperson Yash suspended Shop On : Sunday
Salesperson Yash resumed Shop On : Wednesday
Salesperson Yash Continued Shop On : Thursday
Salesperson Yash Continued Shop On : Friday
Salesperson Yash Continued Shop On : Saturday

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>

```

Program – 37.

Q :- WAP That Read From A File & Write In The File ?

```
import java.io.*;
import java.util.*;
public class Prog37 {
    public static void main(String args[]) {
        try {
            File myFile = new File("Yash.txt");
            myFile.createNewFile();
            FileWriter WriteFile = new FileWriter("Yash.txt");
            WriteFile.write("Hello I Am Yash Pandey\n");
            WriteFile.write("I Am 20 Years Old\n");
            WriteFile.write("I Am In 2nd Year Of B.Tech
                                                                    In MAIT");
            WriteFile.close();
            Scanner sc = new Scanner(myFile);
            while (sc.hasNextLine()) {
                String line = sc.nextLine();
                System.out.println(line);
            }
        }
    }
}
```


Program – 38.

Q :- WAP that illustrates how to process mouse click, enter, exit, press and release events. The background color changes when the mouse is entered, clicked, pressed, released or exited ?

```
import java.awt.*;
import java.awt.event.*;

public class MouseEventsAWT extends Frame implements
                                MouseListener {

    private Panel panel;
    private Label statusLabel;

    public MouseEventsAWT() {
        setTitle("Mouse Events");
        setSize(400, 300);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout());
        panel = new Panel();
        panel.setBackground(Color.WHITE);
        add(panel, BorderLayout.CENTER);
        statusLabel = new Label("No event");
        add(statusLabel, BorderLayout.SOUTH);
    }
}
```

```
        panel.addMouseListener(this);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });
        setVisible(true);
    }

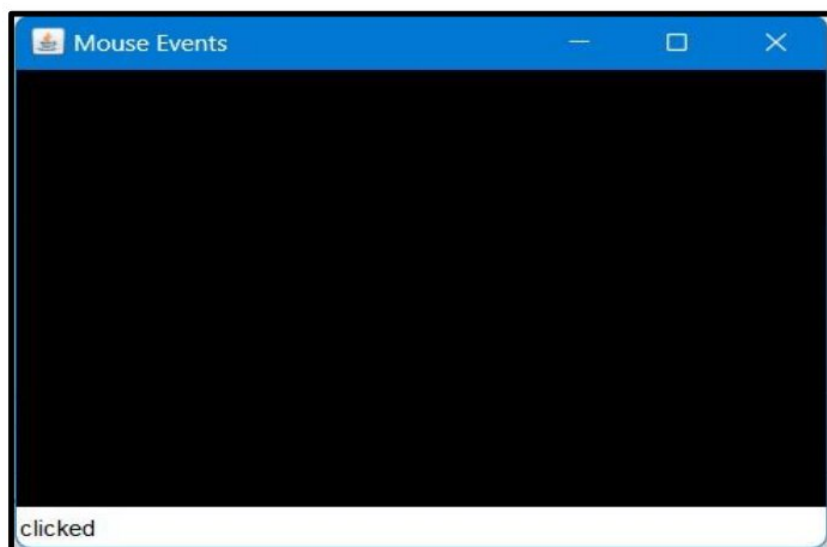
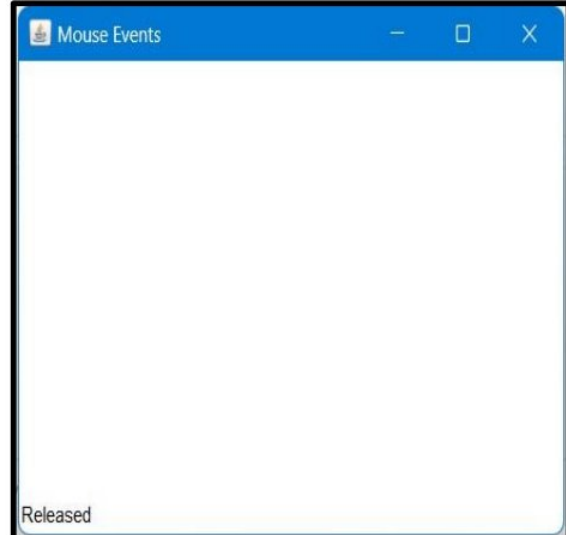
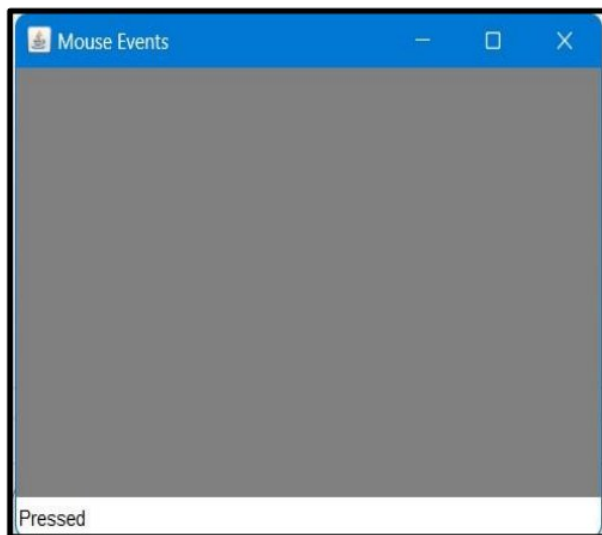
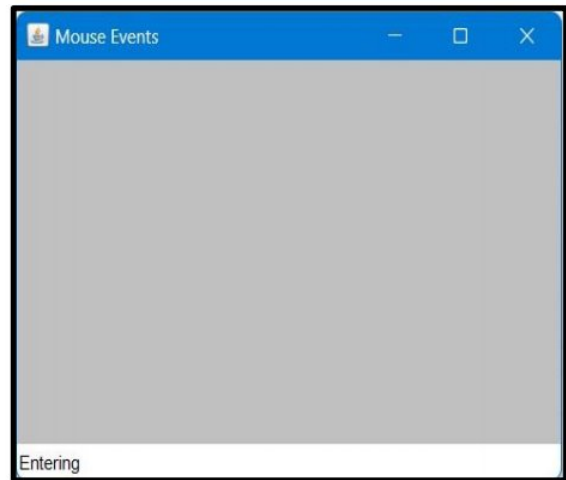
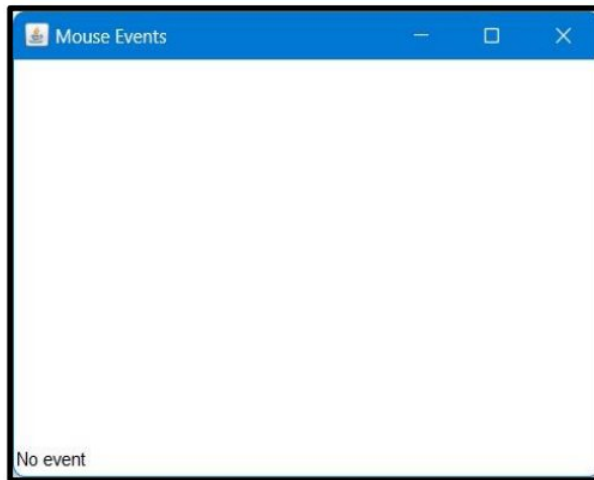
    public void mouseClicked(MouseEvent e) {
        panel.setBackground(Color.BLACK);
        statusLabel.setText("clicked");
    }

    public void mouseEntered(MouseEvent e) {
        panel.setBackground(Color.LIGHT_GRAY);
        statusLabel.setText("Entering");
    }

    public void mouseExited(MouseEvent e) {
        panel.setBackground(Color.WHITE);
        statusLabel.setText("Released");
    }
}
```

```
public void mousePressed(MouseEvent e) {  
    panel.setBackground(Color.GRAY);  
    statusLabel.setText("Pressed");  
}  
  
public void mouseReleased(MouseEvent e) {  
    panel.setBackground(Color.DARK_GRAY);  
    statusLabel.setText("Released");  
}  
  
public static void main( String args[] ) {  
    new MouseEventsAWT();  
}  
  
}
```

OUTPUT :-



Program – 39.

Q :- WAP that displays your name whenever the mouse is clicked.?

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
import java.swing.*;
```

```
public class NameDisplayOnClick extends JFrame {
```

```
    public NameDisplayOnClick() {
```

```
        setTitle("Name Display On Click");
```

```
        setSize(300, 200);
```

```
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
        JPanel panel = new JPanel();
```

```
        JLabel nameLabel = new JLabel("Click Here!");
```

```
        nameLabel.setFont(new Font("Arial", Font.PLAIN, 24));
```

```
        nameLabel.setForeground(Color.BLUE);
```

```
        nameLabel.addMouseListener(new MouseAdapter() {
```

```
            @Override
```

```

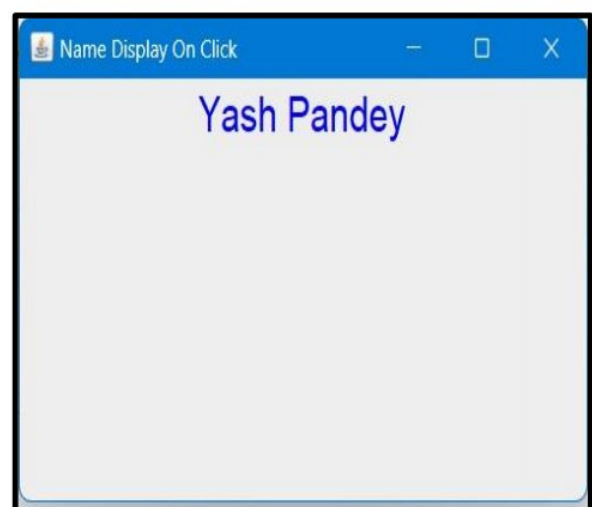
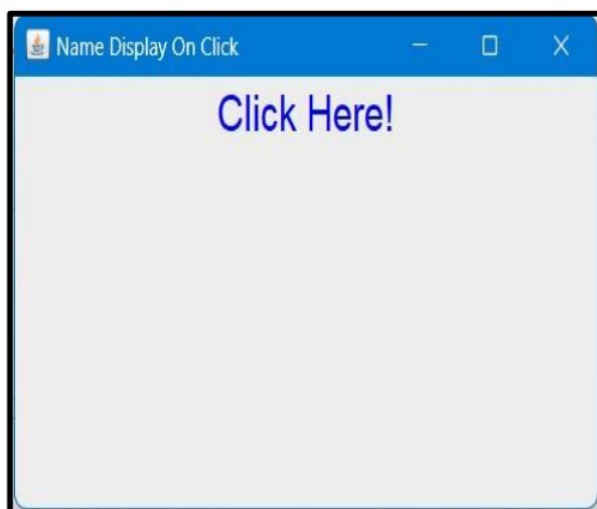
        public void mouseClicked(MouseEvent e) {
            nameLabel.setText("Yash Pandey");
        }
    });

    panel.add(nameLabel);
    getContentPane().add(panel);
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
        NameDisplayOnClick demo = new NameDisplayOnClick();
        demo.setVisible(true);
    });
}
}

```

OUTPUT : -



Program – 40.

Q :- WAP To Convert The Content Of A Given File Into The Uppercase Content Of The Same File ?

```
import java.io.*;
```

```
import java.util.*;
```

```
public class Prog40 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            File MyFile = new File("Yash.txt");
```

```
            MyFile.createNewFile();
```

```
            FileWriter WriteFile = new FileWriter("Yash.txt");
```

```
            WriteFile.write("Hello I Am Yash Pandey\n");
```

```
            WriteFile.write("I Am 20 Years Old\n");
```

```
            WriteFile.write("I Am In 2nd Year Of B.Tech At MAIT");
```

```
            WriteFile.close();
```

```
            System.out.println("Initially , Content In 'Yash.txt' Is ::: ");
```

```
            System.out.println();
```

```
Scanner sc = new Scanner(MyFile);
```

```
while (sc.hasNextLine()) {
```

```
    String line = sc.nextLine();
```

```
    System.out.println(line);
```

```
}
```

```
sc.close();
```

```
FileReader fileReader = new FileReader("Yash.txt");
```

```
BufferedReader bufferedReader = new
```

```
    BufferedReader(fileReader);
```

```
StringBuilder stringBuilder = new StringBuilder();
```

```
while ((line = bufferedReader.readLine()) != null) {
```

```
    stringBuilder.append(line.toUpperCase()).append("\n");
```

```
}
```

```
bufferedReader.close();
```

```
FileWriter fileWriter = new FileWriter("Yash.txt");
```

```
BufferedWriter bufferedWriter = new
```

```
    BufferedWriter(fileWriter);
```

```
bufferedWriter.write(stringBuilder.toString());
```

```
bufferedWriter.close();
```

```
System.out.println("Finally , Content In 'Yash.txt' Is ::: ");
```

```
System.out.println();
```

```
Scanner sc2 = new Scanner(MyFile);
```

```
while (sc2.hasNextLine()) {
```

```
    line = sc2.nextLine();
```

```
    System.out.println(line);
```

```
}
```

```
sc.close();
```

```
MyFile.delete();
```

```
System.out.println("Content converted to uppercase &  
written back to the file successfully.");
```

```
} catch (IOException e) {
```

```
    e.printStackTrace();
```

```
}
```

```
}
```

```
}
```

OUTPUT : -

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SEARCH ERROR

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes\" ;

Initially , Content In 'Yash.txt' Is :::
Hello I Am Yash Pandey
I Am 20 Years Old
I Am In 2nd Year Of B.Tech At MAIT

Finally , Content In 'Yash.txt' Is :::
HELLO I AM YASH PANDEY
I AM 20 YEARS OLD
I AM IN 2ND YEAR OF B.TECH AT MAIT

Content converted to uppercase & written back to the file successfully.

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\Java-LabWork_Codes>
```

➤ The File “Yash.txt” After Code Execution : -

```
J Prog40.java U Yash.txt U X
C++_Codes > Java-LabWork_Codes > Yash.txt
💡 Click here to ask Blackbox to help you code faster
1 HELLO I AM YASH PANDEY
2 I AM 20 YEARS OLD
3 I AM IN 2ND YEAR OF B.TECH AT MAIT
4
```